

Recovering a Hidden Routing Boundary from Query-Only Access

Method note, v6

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August 14, 2026

Abstract

Given query access to a deployed model and a set of real inputs, we estimate the *geometry* of a hidden routing rule. Around each anchor we send a probe cloud shaped by the local data geometry, remove within-branch trend with a trimmed local-linear fit, and test the residual distribution for a two-component structure whose components are linearly separable in input space. From the fitted mixture, its *weight* gives the distance to the boundary and its *separating direction* gives the boundary’s normal; the test is calibrated once, by a parametric Monte-Carlo null run through the full pipeline. Pooling these local estimates across anchors—weighted by evidence rather than thresholded—yields a single hyperplane whose interval demonstrably covers, and, where a bootstrap equivalence bound certifies axis alignment, the coordinate and threshold the system routes on. We characterise where this is identifiable and where it provably is not.

This note supersedes v5 (13 August) by applying the 14–15 August experimental revisions in full: Stage-3 calibration replaced, two of the three Stage-5 checks retired, the pooled estimator repaired, Stage 7 restated as an equivalence claim, interval coverage measured, and the ladder result incorporated. The change-list with evidence is `docs/method_revisions_2026-08-15.md`; the underlying runs are `sim/RESULTS_bootstrap_2026-08-14.md`, `sim/RESULTS_axis_that_2026-08-14` and `sim/RESULTS_ladder_2026-08-15.md`. v5’s structure—organised around the object being estimated, not around a detection statistic—is retained.

Provenance tags used throughout.

[VALIDATED]	measured against ground truth
[SPECIFIED]	designed in detail, not implemented
[NEW]	introduced in v5
[OPEN]	acknowledged gap
[LIMIT]	structural impossibility, not expected to yield to engineering
[RETIRED 14–15 AUG]	stated in v5, falsified by the 14–15 August runs; kept visible, not load-bearing
[REVISED 14–15 AUG]	v5 content corrected by those runs; the corrected form is what stands

What changed from v5, in one box. (1) Stage-3 calibration: the prescribed in-pipeline bootstrap *destroys* the test (null median $0.65 \rightarrow 67.8$); replaced by one parametric Monte-Carlo null through the pipeline, threshold **5.459** at $m = 800$. (2) The dip’s floor is π -dependent— $\approx 6\sigma$ at the method’s own operating point—so the LRT, not the dip, is the detector. (3) The minimum-mass estimability rule is deleted: it passes pure noise at 0.99. (4) The $\hat{\Delta}$ scaling exponent is out of any selection rule, and the 3-rung ladder structurally cannot deliver it at gate anchors (estimable window $2.4 \times$

against a ladder spanning $4\times$); one Experiment A cell is retracted. (5) The pooled offset as specified carries a multiplicative attenuation bias; repaired, its interval coverage goes $0.00 \rightarrow 0.945\text{--}0.998$ at $5\times$ narrower. (6) Axis dominance is restated as an *equivalence* claim with a certified angle bound. (7) Consequence of (2)+(4): the calibrated LRT fires at 1.000 on resonant curvature against 0.203 on the real target, so **the separability guard—never yet run—is now the only confounder defence and sits upstream of everything** (§9, §15).

PART I

What the method claims

1 The target sentence

A deployed model may contain a hidden switch: one rule on some inputs, a different rule on others, with nothing in the interface saying so. We have query access only—no weights, no gradients, no routing metadata, no labels.

The claim we want to end up making is not “*something looks suspicious here.*” It is:

The system routes on coordinate i at threshold $\hat{t} = 4.98 \pm 0.07$, recovered from N anchors under query-only access, $p < 0.05$ against a permutation null.

That is an *inverse problem*, not a test. The question is not “does hidden routing happen”—several published methods answer that. The question is “*given only the system’s outputs, what can I infer about the mechanism that produced them.*” Everything in this note exists to make that sentence defensible.

2 The idea, in words

Pick a real input x —call it an **anchor**. Send a small cloud of nearby queries and look at the answers.

If the anchor sits well inside one routing region, every probe gets the same rule and the answers vary smoothly. If the anchor sits near a boundary, some probes land on the far side and get a *different* rule. Those answers are shifted by a constant—the penalty Δ —so the response distribution splits into two clumps rather than one spread.

Three quantities fall out of that split, and each says something different.

- **How many probes crossed** tells you **how far away the boundary is**. If 5% of probes crossed, the boundary is far relative to the probe radius; if 40% crossed, it is close. The Gaussian geometry converts one into the other exactly.
- **Where the crossers sit** tells you **which way the boundary faces**. The mixture model labels each probe as probably-branch-1 or probably-branch-2. Plot those labels back in input space and fit the separating plane; its perpendicular is the boundary’s direction.
- **Agreement across anchors** tells you **whether they are all seeing the same boundary**, and pooling many noisy local planes gives one global plane with error bars.

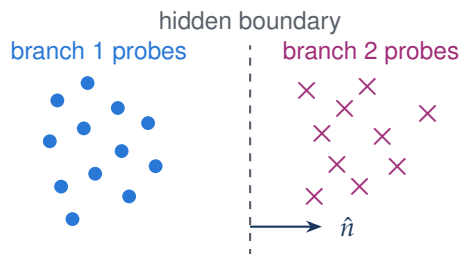


Figure 1: One anchor. The responses say, probabilistically, which side each probe fell on. The plane separating those two groups *in input space* has the boundary's orientation; its perpendicular is $\hat{n}$.

One anchor gives a noisy local estimate. Repeat around many anchors and pool.



Figure 2: Many anchors. Local planes that are individually noisy pool into one global plane. If the pooled normal is dominated by a single coordinate, the rule is axis-aligned and can be stated as a feature and a threshold.

3 A worked example, end to end

Suppose the true hidden rule is $g(x) = \mathbf{1}[x_3 \geq 5]$. You do not know this.

You pick an anchor with $x_3 = 4.7$ and probe with per-coordinate scale $\sigma = 0.5$.

1. The mixture. The two-component fit returns $\hat{\pi} = 0.27$: about 27% of probes landed on the other branch.

2. Distance. The signed displacement along any unit direction is $N(0, \sigma^2)$, so the crossing fraction is $\Phi(-d/\sigma)$. Invert it:

$$\hat{d} = -\sigma \Phi^{-1}(\hat{\pi}) = -0.5 \cdot \Phi^{-1}(0.27) = -0.5 \cdot (-0.613) = 0.31.$$

The true distance is $5 - 4.7 = 0.30$. *You have just measured how far away an unobservable boundary is, from nothing but the proportion of a mixture.*

3. Direction. The probes assigned to the second component are mostly those where x_3 increased. The classifier separating the components returns $\hat{n} \approx (0, 0, 1, 0, \dots)$. Note the distance was *unsigned*—the crossing law gives magnitude only, and the direction comes from $\hat{n}$. You need both.

4. Offset. Anchor at $x_3 = 4.70$, 0.31 away in the $+x_3$ direction: $\hat{c} = 4.70 + 0.31 = 5.01$.

5. Pool. Do this at 100 anchors and combine. You might land at $\hat{n} = (0.01, -0.02, 0.999, 0.01, \dots)$ and $\hat{c} = 4.98 \pm 0.07$ —and the claim in §1 is now supported by evidence.

4 Why the earlier stages exist

Steps 1–5 above are the whole scientific content, and every one of them can be wrong for a reason that has nothing to do with routing. The rest of the pipeline is the list of those reasons and what stops each.

What could go wrong	What stops it
The probe ball never reaches the boundary, so there is nothing to see	Stage 0 — bound the radius by the geometry, and report when no valid radius exists
The probes land where no real input occurs, so the “gate” is real in input space but never fires in deployment	Stage 1 — density filter
The probe cloud inherits the <i>data’s</i> clumpiness and reads as two branches under a perfectly smooth model	Stage 1 — a smooth cloud in the tangent frame is unimodal by construction
A smooth trend across the ball masks the jump—or worse, the fitted plane eats it	Stage 2 — trimmed residualisation
Ordinary curvature produces a lopsided residual distribution that a mixture fits better than one Gaussian	Stage 3 — the separability guard (<i>untested, and since the 14–15 August runs the only defence left standing</i> —§9)
The mixture is real but it is not a <i>boundary</i> —components interleaved in space	Stage 3 — separability again
The local estimates are individually too noisy to trust	Stage 6 — pool, weighting by evidence rather than thresholding

So the stages are not scaffolding around a detector. They are the reasons a reader should believe $(\hat{n}, \hat{c})$.

PART II

The machinery

5 Setup and standing assumptions

$$y = f_{g(x)}(x) + \varepsilon, \quad \varepsilon \sim N(0, \tau^2), \quad g : \mathcal{X} \rightarrow \{1, 2\} \text{ hidden}, \quad g(x) = \mathbf{1}[w^\top x > b].$$

Crossing swaps branch and shifts the output by an additive Δ . Routing is **top-1 and hard**—the entering branch arrives at full weight, so the output *steps*. Top-2 softmax routing instead produces a *kink*: the output stays continuous and only the derivative jumps. That is out of scope, and §11 shows the distinction is detectable rather than assumed.

An anchor set A is drawn from the deployment distribution and **fixed before any query to f** . This is not a convenience: it is what makes the adaptive scale selection in Stage 0 free of any selection penalty. If anchors were chosen using earlier responses, the argument collapses.

6 Stage 0 — where is it legitimate to probe?

Status. Specified, not implemented, and deliberately deferred below the critical path—see §15. The recovery results are established first with the frame and radius *supplied*; this stage is what later lets the method choose them for itself.

Reads anchor covariates only; never touches f . Costs zero queries.

The probe radius must satisfy two things that pull against each other: big enough to reach the boundary, small enough that the honest surface is roughly flat over the ball so the plane fitted in Stage 2 is the right shape.

An intrinsic-dimension curve answers both at once. Estimate $\hat{d}$ at a ladder of neighbour ranks using GRIDE—the local density cancels in a *ratio* of neighbour distances, which is why this family survives clumpy data. The curve has three regimes: inflated at small scales because observation noise is full-dimensional; **flat in the middle—the plateau**; drifting at large scales as the neighbourhood bends or swallows a second cluster. The plateau is the window in which the neighbourhood looks like a flat, evenly-sampled patch.

```
1. NEIGHBORS    N <- k_max nearest neighbours of x in A
                 if too few, or kNN radius too large: ABSTAIN A0_isolated

2. CURVE        for each rank n1 (with n2 = 2*n1):
                 d_hat[n1], CI[n1] <- GRIDE(N, n1, n2)
                 r[n1] <- median distance to the n1-th neighbour

3. PLATEAU      longest contiguous run of rungs whose CIs intersect
                 (j - 1/2, j + 1/2) for a common integer j
                 W <- width of that run, in RANK-decades
                 if W < W_min:
                     if r[k_max] < 3*tau_hat*sqrt(D): ABSTAIN A1n_budget
                     else: ABSTAIN A1_no_plateau

4. SANITY       if d_hat >= gamma*D: FALLBACK to ambient probing

5. FRAME        U <- top d_hat principal directions of neighbours within r_hi
```

Three corrections from the July review are folded in. [\[NEW\]](#)

- The plateau rule is **round-consistency**, not common-intersection. The old rule required all confidence intervals in a run to share a point, which is anti-monotone in data quality: at low noise the intervals are tight, they fail to intersect, and a perfectly flat curve returns “no plateau.” It conflated *statistically indistinguishable* with *geometrically flat*.
- Width is in **rank-decades**, not radius-decades. Since $r \sim k^{1/d}$, a radius-decade threshold is unachievable by construction at $d \geq 5$: ranks 5–200 span 1.6 rank-decades, which is 0.8 radius-decades at $d = 2$ but only 0.32 at $d = 5$.
- **A1 splits from A1n** because they have different remedies. A1 is a property of the anchor—no radius exists at which the assumptions hold, and more budget will not help. A1n is a property of the budget—the ladder never escaped the noise skirt, which extends to roughly $3\text{--}4\tau\sqrt{D}$, not $\sim \tau$. Diagnosable without spending a query.

What Stage 0 cannot tell you: whether the gate is reachable. The plateau is a property of the data’s shape; the gate is

a property of f . Bounding the radius from above makes an out-of-window gate more likely to be missed—the method buys validity partly by shrinking where it looks. [\[LIMIT\]](#)

7 Stage 1 — generating probes that mean something

```
ladder    = S rungs, geometric, from max(r_lo, r_hi/10) up to r_hi    # S = 3
sigma_s   = r_s / sqrt(d_hat)                                     # radius -> per-coordinate scale

z_i       ~ N(0, sigma_s^2 I_{d_hat})                             # in tangent coordinates
delta_i   = U z_i                                                # ambient displacement
keep_i    = DensityFilter(x + delta_i, A)
rho       = |retained| / m
if rho < rho_min: ABSTAIN A3_off_manifold                        # before any query is spent
y_i       = f(x + delta_i)                                       # the only queries spent
```

The scale conversion is not cosmetic. For $z \sim N(0, \sigma^2 I_d)$ the displacement norm concentrates at $\sigma\sqrt{d}$, so placing a rung at radius r requires $\sigma = r/\sqrt{d}$. Getting this wrong reproduces a validated failure mode: under a fixed *total* budget the ball’s radius stops growing with dimension, the crossing rate becomes $\Phi(-d\sqrt{D}/\sigma)$ and collapses, and **detectability**—not merely power—is destroyed. Detectable anchors fell $100 \rightarrow 78 \rightarrow 56$ at $D = 8, 16$. [\[VALIDATED\]](#)

Isotropic in the frame means round within the $\hat{d}$ estimated directions and zero off them. In ambient $\mathbb{R}^D$ that is a flat pancake—highly anisotropic—but isotropic in the coordinates that matter. Roundness matters because the boundary’s orientation is unknown, so a round cloud has no blind spot.

This is also the fix for the failure that killed the tabular experiment. “On-manifold” covers two different operations, and conflating them is fatal:

Sense	Construction	Effect
follows the manifold’s <i>geometry</i>	smooth cloud in the tangent frame	what we want — in-distribution, unimodal by construction
reproduces the data’s <i>density</i>	interpolation / mixup-style re-sampling	inherits the data’s multimodality $\Rightarrow$ false positives

The second sense drove honest false positives to 0.49 in a controlled setting, and up to 1.00 on real building data whose covariates are themselves multimodal (Hartigan’s dip on `YearBuilt`: $p = 0.0000$ —construction booms). A Gaussian cloud in the frame does not resample the data and does not inherit its clustering.

The density filter never sees y . Retention depends on the displacement and the anchor set only, so dropping probes cannot manufacture response-side structure. What it buys is claim scope: reporting $\hat{\pi}$ on *retained* probes is what separates *a gate exists somewhere in input space* from *a gate fires on inputs the model actually sees*. Only the second is an audit claim, and trimming cannot make that distinction.

The ladder, revisited. [REVISED 14–15 AUG] v5 argued $S = 3$ because two rungs cannot separate a gate from resonant curvature ($\hat{\Delta}(r/2)/\hat{\Delta}(r) = 0.94$ resonant vs 1.01 gate; 0.04 at $r/4$). The 15 August ladder experiment showed that argument cannot be cashed at gate anchors. The estimable window in d/σ is bounded below by LTS breakdown ($\pi \lesssim 0.25 \Rightarrow d/\sigma \gtrsim 0.67$) and above by minimum detectable minority mass ($\pi \gtrsim 0.05 \Rightarrow d/\sigma \lesssim 1.64$)—a factor of **2.4**, while a three-rung geometric ladder of ratio 2 spans a factor of **4**. **The ladder does not fit inside the window at any position:** at most two rungs are simultaneously estimable, and only at ratio $\lesssim 1.55$ (untested). Re-anchoring was also tested and made every axis worse—gate detection falls monotonically while every resonant surface reaches fire rate 1.000—so the current position ($\kappa = 0.78$, top rung on the orientation-useful shell) stands. The multi-rung structure is retained for the distance-constancy check (§11b, itself now flagged) and for gap estimation, not as a curvature defence.

8 Stage 2 — keeping the fit off the minority branch

Least trimmed squares with $h = 0.75m$, fitted in tangent coordinates:

1. fit $y \sim a + b'z$ on all retained probes
2. drop the worst-fitting 25% by $|residual|$
3. refit on the remaining 75%
4. residualise ALL points against the refit plane

The mechanism is not variance reduction—it is **keeping the fitted plane on one branch**. Plain OLS finds a compromise plane tilted between the two branches and absorbs

$$\Delta \left(1 - 2\varphi(0)\sqrt{2/\pi}\right) \approx 0.36 \Delta,$$

about **64% of the gap** at balanced mixing. Trimming discards points that are predominantly the minority branch, so the plane commits to the majority and the jump reappears at full size.

Evidence: power flat in ambient dimension at 0.74–0.76; clumpy-manifold honest FP 0.49 $\rightarrow$ 0.00; trend-hidden gaps unmasked 0.14 $\rightarrow$ 0.90. [VALIDATED]

Structural cost: 25% breakdown by construction, so $\pi \approx 0.5$ is out of reach—and at exactly 0.5, “majority branch” stops being a defined concept. [LIMIT]

9 Stage 3 — is this mixture attributable to a boundary?

Two tests in parallel at each rung, with a guard.

$$H_0 : r \sim N(\mu, \sigma^2) \quad \text{versus} \quad H_1 : r \sim \pi N(\mu_1, \sigma^2) + (1 - \pi) N(\mu_2, \sigma^2)$$

with σ shared between components.

The dip tests only “the density has one hump.” It assumes almost nothing and stays valid under curvature. [REVISED 14–15 AUG] But its floor is a function of π , not the flat “ ~ 2 standard deviations” v5 stated (that is the equal-weight case). Separation required for power 0.5 at $n = 800$:

π	0.50	0.35	0.25	0.10	0.05
separation	3σ	4σ	4σ	6σ	$> 7\sigma$

Anchor placement targets $\pi \leq 0.10$, so at the method’s own operating point the dip is inert below $\approx 6\sigma$ —against a real scaffold at $1.6\text{--}1.95\sigma$. Calibrating it through the pipeline restores nominal size but buys *no* power and raises resonant-curvature false positives from 0.058 to 0.557. **Keep the dip on its tabled p -value as a conservative side-check; the LRT is the detector.**

The equal-variance LRT buys the sub-bimodal regime: 0.42–0.54 at 1.5 sd, 1.00 at 2.5 sd. The shared σ matters technically—an unrestricted normal-mixture LRT diverges—and statistically, since a narrower alternative has more power. It is close to efficient rather than merely better: the best single Gaussian matches the mixture’s first two moments exactly, so all signal lives in skewness and excess kurtosis, and power predicted from moment structure (0.55/0.30 at $\pi = 0.25/0.50$) matched observation (0.54/0.32).

Calibration: one parametric Monte-Carlo null through the pipeline. [REVISED 14–15 AUG] v5 prescribed an in-pipeline bootstrap of inlier residuals and warned against clean-Gaussian thresholds. Both halves were wrong, in opposite directions. Resampling *inlier* residuals inflates the null LRT from a true median of 0.65 to **67.8**—the inlier set is the truncated middle 75% (kurtosis 2.11 against the noise’s 2.99) and the pipeline truncates it again, so every p -value is 1.000 on honest and gated surfaces alike. And the deflation premise is untrue: Stage 2 residualises *all* points, so the trim selects the *fit*, not the test sample. Pipeline null $q_{95} = 5.459$ against untrimmed clean-Gaussian 5.543 ($B = 20,000$ each).

What stands instead: **one parametric Monte-Carlo null through the full pipeline, computed once at high B —not a bootstrap, and not per anchor.** The null is invariant to rung, τ , trend, and anchor (per-anchor variation at $B = 300$ is pure Monte-Carlo noise, sd 0.51 vs 0.50 for pure resampling). The one real dependency is sample size—5.62 at $m = 400$, 5.459 at $m = 800$, 4.99 at $m = 2000$ —so **calibrate at the deployed m .** (The July threshold of 6.08 is reproduced exactly by clean draws at $m = 400$: a sample-size difference, not a trimming effect.) The threshold holds up: across the entire ladder sweep the honest surface fires at 0.028–0.052 against nominal 0.05. Costs zero model queries.

The separability guard—and this is where the estimator is born. A real gate is not merely a mixture in response space. It is a mixture whose components are **linearly separable in z** , because the boundary is locally a hyperplane; curvature artifacts produce components interleaved in space. So fit a linear classifier on (probe coordinates, responsibilities) and check held-out balanced accuracy: near-perfect for a gate, near chance for an artifact.

This closes the LRT’s one hole—and **it is the only thing left that can.** The dip’s null is broad and survives curvature; the LRT’s null is “one Gaussian,” and curvature gives skewed-but-unimodal residuals that a mixture fits better. The exposure is now measured, not hypothetical: **the calibrated LRT fires at 1.000 on resonant curvature against 0.203 on the real target**, so a screen built on it selects confounders $5\times$ more readily than gates. The 14–15 August runs eliminated every other instrument that was meant to cover this (the scaling exponent, §11a; the minimum-mass rule; the dip’s calibrated variant), which leaves every flag conditioned on this guard—and **the guard has never been run.** It is the next experiment, upstream of any screening, selection, or pooling; its design, pre-registered predictions, and stop/go criteria are in `review/TODO.md`.

And the guard’s decision boundary, lifted through the frame U , is the estimate of the gate’s normal. Test and recovery are the same computation. This is the hinge of the whole note.

PART III

The estimator

10 Stage 4 — what one anchor knows

Five outputs per anchor per rung. Three exist; two are new.

Gap. $\hat{\Delta} = |\hat{\mu}_1 - \hat{\mu}_2|$ from the mixture fit. Recovered 0.291 against a planted 0.300, honest-model baseline 0.014. [VALIDATED]

Composition. $\hat{\pi}$ from the same fit, oriented query-only: the anchor’s own response says which component it sits in, and the crossing mass is the other one. [VALIDATED]

Distance. [NEW] For a unit normal and probes $z \sim N(0, \sigma_s^2 I)$, the signed projection is $N(0, \sigma_s^2)$; crossing means the projection exceeds d . Hence

$$\hat{d}_s = -\sigma_s \Phi^{-1}(\hat{\pi}_s)$$

the crossing law read backwards. The law itself is [VALIDATED]—verified to three decimals in both its fixed-per-coordinate and fixed-total forms. Using it as an *estimator* is new, and as of 13 August it is measured:

1-D known-regimes setting. $\Delta = 0.30$, ladder `geomspace(0.02, 0.2, 3)`, $m = 1000$ per rung, with two query-only guards: the dip fires under Bonferroni, and $\hat{\pi} \in [0.05, 0.50]$. $\text{corr}(\hat{d}, d_{\text{true}}) = 0.9958$ ($n = 217$) · median relative error -0.92% · median absolute error 0.0031

Tracks correctly from $d = 0.012$ to 0.251, and is flat across a $10\times$ noise sweep: median absolute error 0.0029/0.0031/0.0027 at $\tau = 0.005/0.02/0.05$.

Two cautions carried by that result. Relative error degrades near the boundary—3.3% beyond $d = 0.1$, 5.0% at 0.05–0.1, 26% below 0.01—where the *absolute* error is still ~ 0.003 and only the denominator is small. Report absolute error near the boundary.

And **the estimate is conditional on detection**. Without the dip guard, anchors beyond the ladder top return confident nonsense (true 0.277 against estimated 0.141, with enormous spread), because $\hat{\pi}$ there is not a crossing fraction at all. The guard is query-only, so this is legitimate—but the conditioning belongs in the claim.

[REVISED 14–15 AUG] “Estimable” is defined by the calibrated test, not by a minimum-mass rule. v5’s implicit criterion— $\hat{\pi} \geq 0.05$ defines the deepest usable rung—is not a gate: on a surface with *no gate anywhere* it passes at 0.990–0.995 (EM happily splits pure noise), and it is anti-correlated

with the truth, rejecting rungs where EM has found a real but tiny minority at 0.748. Deleted as an estimability criterion. $\hat{\pi}$ bounds are retained for exactly one purpose: keeping LTS away from its breakdown point.

Direction. [VALIDATED] The separability classifier’s boundary, lifted to ambient space:

$$n_a = \frac{Uv}{\|Uv\|} \quad (\text{oriented: higher-mean component on the } + \text{ side}), \quad c_a = n_a^\top x_a + \frac{b}{\|v\|}.$$

Measured 13 August at intrinsic $d = 2$, ambient $D = 20$, frame supplied, 200 anchors per cell over a $\Delta/\tau \times \pi$ grid. The concern this had to answer is that responsibilities are *estimated*, not true branch labels, so the chain misassignment $\rightarrow$ classifier label noise $\rightarrow$ bias in $\hat{n}$ could dominate everything else near weak separation. It does not:

Misassignment costs variance, not bias. Mean signed rotation per cell: the oracle arm (true branch labels) lies in $[-0.35^\circ, +0.27^\circ]$; the estimated-responsibility arm lies in $[-6.6^\circ, +5.9^\circ]$ and is **not significant at 95% in any of the 20 cells**.

This is the symmetry argument holding: the responsibilities are a function of the one-dimensional residual, the residual is a function of $z^\top n$, and so the class-conditional means of z remain on n however badly the labels are corrupted. Corrupted labels *attenuate* the separation; they do not rotate it.

Because bias is what survives pooling and variance is not, this is the outcome that matters. Pooled orientation error falls as $1/\sqrt{N}$: at $\pi = 0.05$ it goes $6.8^\circ \rightarrow 0.9^\circ$ between $N = 1$ and $N = 100$ at $\Delta/\tau = 2.5$, and $20.4^\circ \rightarrow 1.8^\circ$ at $\Delta/\tau = 1.5$ — which is below the dip’s floor entirely.

Fit the discriminant on all probes, not on the trimmed inliers. [NEW] Median orientation error in the informative cells is 5.8° using every probe against 21.9° using the LTS inlier set. The trim removes points that are predominantly the minority branch — which is exactly the branch carrying the crossing signal. So: trim to obtain clean residuals for the *mixture*, then discard the mask for the *discriminant*. Soft responsibilities also beat hard $\gamma > 0.5$ labels (5.8° against 7.6°); keep the uncertainty.

Where the useful anchors are, and it is not where one would guess. [NEW] Per-anchor orientation error is **monotone increasing in π** , not U-shaped: at $\Delta/\tau = 5$ the per-anchor standard deviation runs 3.9° at $\pi = 0.05$ to 47.5° at $\pi = 0.50$. Fewer crossers, but they sit much further out along the normal, and the separation of class-conditional means along n *rises* as π falls:

π	0.05	0.10	0.20	0.35	0.50
mean separation along n	2.17σ	1.95σ	1.75σ	1.63σ	1.60σ

At $\pi \rightarrow 0.5$ there is additionally no majority branch for LTS to commit to, so the plane lands between branches and corrupts the residuals the mixture sees. The useful anchors sit at $d \approx 1.3\text{--}1.6\sigma$. This agrees with the distance estimator, which is also most accurate at larger d/σ , so a single rung serves both.

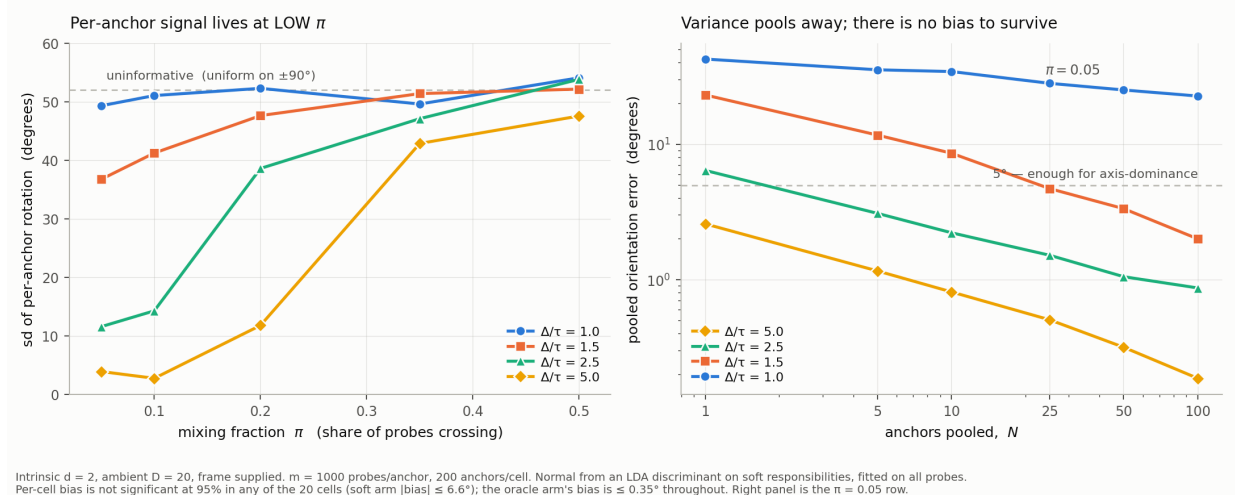


Figure 3: Left: per-anchor information against the mixing fraction. Signal lives at *low* π ; 52° is what uniformly random directions give. Right: pooled orientation error against the number of anchors, at $\pi = 0.05$.

Weight. w_a is the LRT statistic, **not thresholded**. At per-anchor power near 0.5, thresholding before pooling discards half the true signal.

Every anchor emits $(n_a, c_a, \hat{d}_a, \hat{\Delta}_a, w_a)$ —including anchors that did not fire.

11 Stage 5 — the consistency checks, after the 14–15 August runs

v5 offered three free checks on queries already spent. The runs retired (a) and (c) and flagged (b). This section keeps all three visible—the retirements changed the critical path (§15) and should not be re-proposed.

(a) How the response scales with radius [RETIRED 14–15 AUG]

Regress $\log \hat{\Delta}$ on $\log r$ across rungs; report the exponent α .

Structure	$\hat{\Delta}$ scaling	ratio at $r/2$	ratio at $r/4$
step (top-1 gate)	r^0	1.00	1.00
kink (top-2 softmax)	r^1	0.50	0.25
curvature (confound)	r^2	0.25	0.06

The table stands as *characterisation*—the physics of steps, kinks, and curvature is right, and early observations matched it (gates 1.01; resonant 0.04 at $r/4$ against r^2 's 0.0625). What died is its use in any *selection rule*, for two independent reasons. First, availability is inverted: α is fittable at **17%** of gate anchors and **100%** of curvature anchors—a discriminator that cannot be computed at the anchors you want to keep is not a filter. Second, it stops discriminating: the r^2 law holds only while the probe ball is small relative to the curvature wavelength; a wide ball spanning many periods sees the sine's fixed full amplitude, which is scale-free, and resonant α collapses to ≈ 0.00 . The associated Experiment A cell ($\pi_{\text{top}} = 0.35$, " r^0 recovered exactly") is **retracted**: an honest surface with no gate reproduces it exactly (three "estimable" rungs, median $\alpha = -0.052$, closer to

zero than the real gate's -0.160)—the exact recovery was measuring noise's scale-freedom, via the minimum-mass rule passing pure noise. At most, α survives as a population-level diagnostic in the small-ball regime, which is the regime the ladder cannot reach and stay estimable.

(b) Whether the geometry is self-consistent [NEW] [OPEN]

For a planar boundary at fixed distance, $\hat{d}_s$ should not depend on the rung. A gate returns the same d at every scale; a smooth confound's $\hat{\pi}_s$ is a mixture fit to something that is not a crossing, so its implied distance wanders. Measured 13 August:

Model	median CV of $\hat{d}$	fire rate
gated	0.098 (0.074 at $\tau = 0.02$)	0.678
GP null, $\ell = 0.05$, $A = 0.10$	0.557	0.397
GP null, $\ell = 0.05$, $A = 0.20$	1.042	0.114
kink control	1.241	0.023
honest	—	0.000
GP null, $\ell = 0.30$ (above ladder top)	—	0.000

At $CV < 0.15$: **56% of gates retained, 93% of confounders rejected**. [REVISED 14–15 AUG] Flagged for re-examination rather than retired: the check needs ≥ 2 estimable rungs at one anchor, and the ladder result says the estimable window (factor 2.4) admits at most two rungs only at ratio $\lesssim 1.55$. The $CV = 0.098$ figure above was measured under the 1-D naive protocol with a different geometry; do not carry it into the current design without re-measurement.

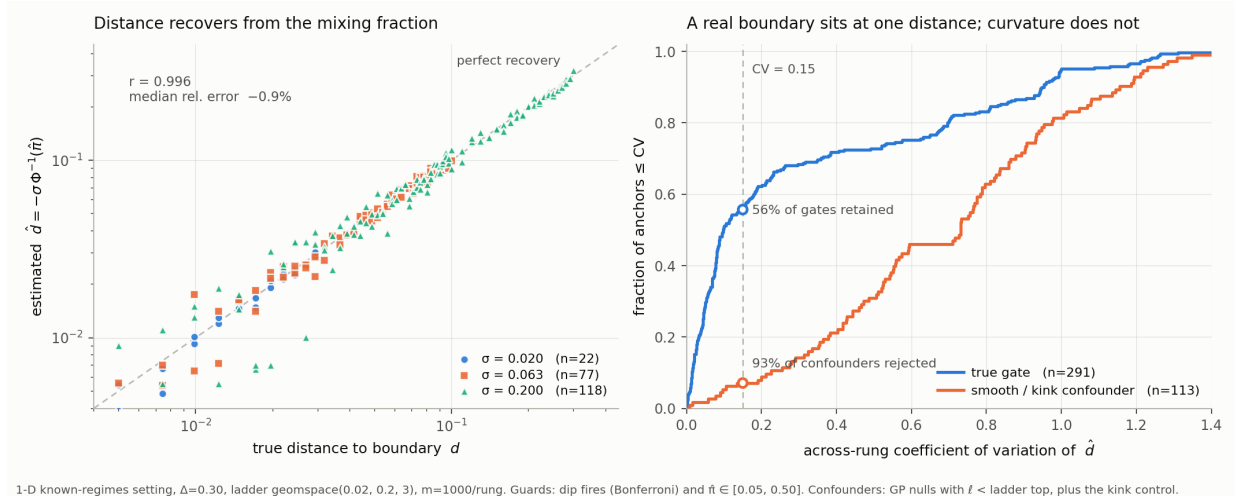


Figure 4: Left: the distance estimator against ground truth, by probe scale. Right: across-rung constancy of $\hat{d}$ separates a true gate from smooth and kink confounders.

(c) Whether the two routes to the offset agree [RETIRED 14–15 AUG]

The offset is available twice: from the classifier's own intercept $b/\|v\|$, and from $-\sigma\Phi^{-1}(\hat{\pi})$ signed by $\hat{n}$. v5 claimed they "agree by construction under a correctly specified model," making their gap a misspecification alarm. **They do not agree, and one of the arms is biased.** They diverge systematically whenever $\pi \neq 0.5$: the LDA midpoint of the projected class means sits at

0.780σ against a true 1.282σ at $\pi = 0.10$ —for a half-space cut of a Gaussian the two conditional means are $-\sigma\varphi/\Phi$ and $+\sigma\varphi/(1 - \Phi)$, whose midpoint is not the cut distance unless the split is balanced. The comparison was detecting a real defect in its own classifier arm. Retired: **the crossing-law route is the estimator** (it is the validated one), and the agreement check goes with the biased arm. If a specification diagnostic is wanted here, it must be re-derived after the §12 repair.

Structure that fires the LRT but fails the separability guard routes to `A5_structure_unattributed`—structure present, not attributable to a boundary. A third outcome: not a flag, and not a clean bill.

12 Stage 6 — pooling

[VALIDATED] (Experiment P, 13 August; offset estimator repaired 14 August.) A boundary is a shared object: one hyperplane passes through many probe balls. Per-anchor power of 0.5 barely matters if evidence for a *common* boundary aggregates over 200 anchors—which means **per-anchor power is the wrong figure of merit**. The right one is the number of anchors needed to recover the boundary.

Do not threshold before pooling. All anchors contribute, weighted by w_a .

	Estimator	Uses geometry
A	Fisher/Stouffer combination of per-anchor p -values	no — baseline
B	threshold on w_a , then cluster (n, c)	partly
C	w -weighted mode-seeking (mean-shift / Hough) over all anchors	fully

Measured: $C > B > A$ at every Δ , **largest gap** at $\Delta/\tau = 1.5$ —confirmed as predicted, and exactly where the real target sits. Two conditions arrived with the confirmation.

Placement is a prerequisite, not a nicety. Geometric pooling recovers the boundary at **0.96** with anchors placed by design near it, and **0.00**—not slower, dead—with anchors placed at random. Boundary-seeking anchor placement does not exist yet, and every pooling number in this note is conditional on it. [OPEN]

The pooled offset, repaired. [REVISED 14–15 AUG] As specified, each anchor’s offset used *that anchor’s own* normal against a position far from the origin: writing the anchor as $t_a = (T \pm d) \nu_{\text{true}} + \text{tangent}$, the pooled term picks up a factor $\mathbb{E}[\cos \varphi_a] < 1$ from per-anchor orientation error φ_a . The bias is **multiplicative in the true threshold** and does not shrink with N (measured bias/ T : -0.29 against the predicted $\mathbb{E}[\cos \varphi] - 1 = -0.26$ at $\Delta/\tau = 1.5$). Experiment P could not see it because P planted $c^* = 0$, where a multiplicative bias is invisible by construction—an *estimator validated only at the origin has not been validated for a threshold*. A second, additive defect: the LDA-midpoint offset sits at 0.780σ against a true 1.282σ at $\pi = 0.10$ (§11c). The repair, in two parts: **project each anchor’s estimated boundary point onto the pooled normal** (per-anchor rotation error is then no longer multiplied by the anchor’s distance from the origin), and **take anchor-to-boundary distance from the crossing law**, not the LDA midpoint—direction from separability, distance from the crossing law, each instrument where it is validated. Not a trade-off: coverage $0.945\text{--}0.998$ and the interval is $5\times$ narrower (0.127 vs 0.691 at $\Delta/\tau = 1.5$, $N = 200$).

A no-information guard is required. [NEW] Outside the operating region — $\Delta/\tau = 1.0$ at any π , or $\pi \gtrsim 0.35$ — the per-anchor rotation has standard deviation $\approx 52^\circ$, which is exactly what uniformly random directions give. Pooling *axial* estimates that carry no information does not converge to the truth and does not converge to nothing: the mean direction of N random axes is itself uniformly random, so the pooled estimate lands on an arbitrary direction with an interval that tightens around it. **Pooling can therefore be confidently wrong in a way that per-anchor abstention never is.** Stage 6 needs an explicit check that the anchor set carries information before it reports — the permutation null is the natural instrument, since uninformative input should be indistinguishable from permuted input.

Null. Permute the residual-to-point assignment within each anchor and rerun the pipeline. This randomises normals while preserving marginal residual distributions; ≥ 500 permutations.

Metrics. $N \mapsto$ orientation error, $N \mapsto |\hat{c} - c^*|$, and N_{95} . The first two are the estimation curves; N_{95} is the detection curve. Under this note’s framing, the first two are the headline.

The registered warning fired. [RETIRED 14–15 AUG] v5 registered, before running: the naive argument for pooling—real boundaries give consistent normals, artifacts give random ones—**fails for the confound that matters**, since a surface with smooth directional structure has a preferred direction; the hoped-for rescue was that gate offsets form a single mode in c while curvature’s form a comb. **The prediction of failure was correct and the rescue was not** (P-2, falsified): long-wavelength curvature pools *coherently* into a 100%-confident false boundary with direction coherence $R \approx 0.879$ on a surface containing no boundary at all. Pooling therefore cannot reject smooth confounders on its own, and—per the registered contingency—smooth rejection returns to the per-anchor rules. After the 14–15 August retirements in §11, the per-anchor rule that remains is the separability guard (§9), which must be tested *at this pooled level too*, not only per anchor.

13 Stage 7 — stating the rule

[REVISED 14–15 AUG] v5’s rule—“is one $|\hat{n}_i|$ dominant? calibrate against the permutation distribution of $\max_j |\hat{n}_j|$ ”—is structurally broken, in a way worth recording because the repair reframes what kind of claim this is. $\hat{n}$ always lies in $\text{span}(U)$, and permuting residuals *within* an anchor leaves the frame untouched, while an axis-aligned gate that is visible at all forces the frame to contain that axis; the permuted normal is then $\cos \varphi u_1 + \sin \varphi u_2$ and the null of $\max_j |\hat{n}_j|$ has $q_{95} = \sin(0.95 \pi/2) = 0.9969$. The rule therefore reduces to “is the pooled orientation error below $\approx 4.5^\circ$,” which is not a test of axis alignment: on oblique gates, where any single-coordinate claim is *wrong* rather than weak, it fires at 0.20–0.79. (A bootstrap-stability repair was tested and is worse— ~ 1.00 at every tilt—because the argmax of a well-determined oblique normal is *stably* the largest component. **Stability is not validity.**)

Axis dominance is an equivalence claim, not a significance claim. The burden is not “the normal beats a no-signal null”—a normal can beat that null by being merely well determined while pointing anywhere. The burden is that the off-axis component is *provably small*.

```

bootstrap the anchors; form a one-sided UPPER confidence bound on
angle(n_hat, candidate axis)

if bound < stated tolerance:  claim coordinate i,
                              threshold t_hat = c_hat / n_hat_i,
                              interval from the Stage-6 repaired estimator
else:                          report the hyperplane (n_hat, c_hat)
                              AND the bound itself

```

The bound is an honest estimate of the tilt—certified $0.87^\circ/10.14^\circ/20.13^\circ/30.09^\circ/45.12^\circ$ against true $0/10/20/30/45$ —and the false-claim rate beyond tolerance is 0.000. The tolerance is an explicit, reportable scientific choice, argued from the deployment rather than hidden in a calibration. **Report the bound, not the binary:** at $\Delta/\tau = 2.5$, 25 anchors certify better than 7° and the target sentence is available; at $\Delta/\tau = 1.5$ —the real scaffold’s regime—200 anchors certify only $\approx 13.5^\circ$, and the honest output is the hyperplane plus the bound; at $\Delta/\tau = 1.0$ no feature can ever be named.

PART IV

What must still be true

14 The obligation the reframe creates

A detector is validated by power and false-positive rate. **An estimator is validated by bias, variance, interval coverage, and query complexity.** Coverage matters most: the ± 0.07 in the target sentence is what turns a point estimate into a scientific claim.

[REVISED 14–15 AUG] Measured, 14 August—and the specification failed exactly here. Coverage of the nominal 95% interval on $\hat{t}$ as specified: **0.00** at $N = 100$ –200 for $\Delta/\tau \leq 2.5$, and it gets *worse* as N grows—the signature of bias plus a shrinking interval tightening around a displaced centre. Even at $\Delta/\tau = 5.0$ it plateaus at 0.93, never reaching nominal. After the §12 repair: **0.945–0.998**, with a $5\times$ narrower interval. Scope of the measurement, stated plainly: $\hat{t}$ only, axis-aligned gate ($\theta = 0$), by-design anchor placement, Gaussian noise, intrinsic $d = 2$, frame supplied.

Calibrated coverage of every interval the method reports is a first-class deliverable, not a finishing touch. $\hat{t}$ now has it; the certified angle bound has it by construction; anything else the paper prints an interval on must measure it first.

15 The critical path

1. **Validate $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$.** **Done 13 August** (§10).
2. **Validate the separability-derived normal.** **Done 13 August** (§10): no detectable bias in any cell, and pooled error below 3° at $N = 100$ down to $\Delta/\tau = 1.5$. Intrinsic $d = 2$ only — the sweep over $d \in \{4, 8\}$ is a parameter change in the same harness and remains to be run.
3. **Experiment P.** **Done 13 August:** $C > B > A$ confirmed with the largest gap at $\Delta/\tau = 1.5$; placement shown to be a prerequisite (0.96 by design vs 0.00 random); P-2 falsified — pooling does not reject smooth confounders (§12).

4. **Axis-aligned threshold recovery with a *calibrated* interval.** Done 14 August, after two repairs (§12, §13, §14).
5. **The separability guard. *Untested, and now load-bearing.*** The ladder result removed the scaling exponent from the selection rule, and screening on the calibrated LRT enriches curvature $5\times$ against the real target (1.000 resonant vs 0.203). The guard is the only remaining instrument that distinguishes a boundary from curvature, per anchor *and* at the pooled level. It moved onto the critical path *ahead of* the screening experiment, not after it.
6. **screen $\rightarrow$ select $\rightarrow$ recover, end to end.** Not started—the binding item, gated on the guard’s verdict.

On deferring Stage 0. The order above proves geometry recovery first under *known-valid* local probes—a flat manifold of known dimension, where the frame and radius can be supplied rather than estimated—and adds automatic geometry selection afterwards. This is deliberate: the contribution is that $(\hat{n}, \hat{c})$ is recoverable at all, and establishing that under a hand-supplied probe geometry isolates it from any question about how the geometry was chosen. Stage 0 is then the extension that *removes* the assumption, converting “given a valid local probe geometry” into a condition the method checks for itself and declines when it fails. Deferred, not dropped: §6 remains the specification, and the results in steps 2–4 must state the conditioning explicitly. The one part of Stage 0/1 kept regardless is the **density filter**, because it scopes the claim rather than improving it.

Where the error budget lives — answered. *The prediction below was made before step 2 and is retained with its outcome.* Responsibilities are *estimated*, not true branch labels. The chain runs: mixture misassignment $\rightarrow$ classifier label noise $\rightarrow$ bias in $\hat{n}$. Near weak separation the responsibilities go diffuse ($\gamma \approx 0.5$ almost everywhere) and the discriminant direction destabilises; this may dominate finite-sample error entirely in the regime that matters. The clean way to isolate it is to **fit the classifier twice**—once on estimated responsibilities, once on true labels—and difference the orientation errors, as a function of Δ/τ and π , not only N . If that gap is large at $\Delta = 1.5$ sd, pooling must weight by *assignment confidence* rather than by the LRT statistic alone.

Outcome: the gap is entirely variance. Bias is not significant in any cell, so weighting by the LRT statistic is not disqualified, and the binding constraint turns out to be anchor *placement* ($\pi \leq 0.10$) rather than the weighting rule.

16 Scope: this is a locally single-boundary method

The procedure identifies **at most one boundary per probe ball**. Its precondition is $r_{hi} < L$, where L is the routing cell scale—and that precondition **is not checkable from the covariates alone**, because routing density lives in the response while covariate geometry is unchanged. [LIMIT]

When it is violated—many experts, small cells, a ball spanning several—the many small-separation components read as heteroscedastic noise, and **Stage 0 is blind. No abstention fires.** This is the only failure mode in the method that returns a confident wrong answer rather than declining to answer.

Two things follow. It explains why $K = 2$ globally is the right scope rather than an arbitrary simplification: the local procedure never sees more than one boundary regardless, so global K must be *assembled* from local evidence, never fitted as a large mixture. And it suggests a response-side detector that does not yet exist—if a ball spans several cells, $\hat{\Delta}$ may shrink with r rather than staying flat, which the existing ladder would see. Worth testing in the step-2 harness; not claimed.

[OPEN]

17 Everything else that limits the claim

- **Parallel penalty.** An offset aligned with strong within-branch variation is invisible at every radius, in every dimension, under every projection. Proven, not conjectured. [LIMIT]
- **Reach is not validity.** The plateau says where assumptions hold, not where the gate lives. [LIMIT]
- $\pi \approx 0.5$. LTS breakdown is structural. The LRT fixes *testing* (0.32, graceful); recovery still breaks, since there is no majority branch to trim toward. [LIMIT]
- **Detection floor** ≈ 1.5 sd at $m \sim 10^3$. Moment z-scores scale like $\sqrt{n}$, so halving the detectable separation costs $4\times$ the queries.
- **Coverage.** Randomly placed anchors contain a boundary about r_{hi}/L of the time—roughly 15–20%. Experiment P upgraded this from a constraint to a prerequisite: geometric pooling is 0.96 with by-design placement and 0.00 with random placement. Boundary-seeking placement does not exist and is now on the critical path for any recovery claim. [OPEN]
- **The honest reach at the real target’s signal.** At $\Delta/\tau = 1.5$ —the housing scaffold’s measured regime—200 anchors certify the routing direction only to $\approx 13.5^\circ$, so a single-coordinate claim is not available there; naming a feature needs $\Delta/\tau \geq 2.5$. Not a defect: the honest reach of query-only access against a router that hides in the noise, to be stated rather than hidden.
- **Validation scope.** One model class, one D , one τ_y ; frame supplied throughout the estimator runs. The “no full-pipeline bootstrap yet” entry is closed—the answer is that it should never be a bootstrap (§9): one parametric Monte-Carlo null through the pipeline, computed once. [REVISED 14–15 AUG]

18 Provenance summary

Component	Status
Crossing law; fixed-per-coordinate budget mandatory	[VALIDATED] to three decimals
Trimmed residualisation, dimension-independent power	[VALIDATED] 0.74–0.76 flat in D
OLS absorbs $\sim 64\%$ of the gap	closed form, matches measurement
Trimming restores validity on clumpy manifolds	[VALIDATED] $0.49 \rightarrow 0.00$
Dispersion cannot discriminate a gate from curvature	[LIMIT]
Dip floor is π -dependent ($\approx 6\sigma$ at $\pi \leq 0.10$); LRT is the detector	[REVISED 14–15 AUG] measured
LRT calibration: parametric MC null through pipeline, 5.459 at $m = 800$	[VALIDATED] honest fire 0.028–0.052

Component	Status (cont.)
In-pipeline bootstrap calibration (v5 prescription)	[RETIRED 14–15 AUG] inflates null 0.65 → 67.8
Minimum-mass estimability rule	[RETIRED 14–15 AUG] passes pure noise at 0.99
Local $\hat{\Delta}$ recovery	[VALIDATED] 0.291 vs 0.300
Group-conditional estimand at the anchor	[VALIDATED] cosine 1.000
Distance estimator $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$	[VALIDATED] $r = 0.996$, median rel. error -0.9%
Distance invariance across rungs	[REVISED 14–15 AUG] re-examine: measured under a different geometry; ladder window admits ≤ 2 rungs
$\hat{\Delta}$ scaling exponent in any selection rule	[RETIRED 14–15 AUG] fittable at 17% of gates, 100% of confounders; Exp. A $\pi_{\text{top}}=0.35$ cell retracted
Offset-agreement check	[RETIRED 14–15 AUG] its classifier arm is biased (0.780σ vs 1.282σ)
Round-consistency plateau; rank-decade W ; A1/A1n split	[NEW], unimplemented
Stage 0 entire (GRIDE, plateau, frame)	[SPECIFIED] — no implementation exists
Boundary normal from the separability classifier	[VALIDATED] no bias in 20/20 cells
Pooled orientation error $< 3^\circ$ at $N = 100$, $\Delta/\tau \geq 1.5$	[VALIDATED] (at $d = 2$)
Discriminant on all probes, not LTS inliers	[NEW] 5.8° vs 21.9°
Orientation error monotone increasing in π	[VALIDATED] — prediction of a U-shape falsified
No-information guard for pooling	[NEW] — required, not yet designed
Intrinsic-dimension scaling of orientation error	[OPEN] — $d = 2$ only
Pooling: $C > B > A$, largest gap at $\Delta/\tau = 1.5$	[VALIDATED] Experiment P
Pooling requires by-design placement	[VALIDATED] 0.96 vs 0.00 — placement rule [OPEN]
Pooling rejects smooth confounders on its own (P-2)	[RETIRED 14–15 AUG] pools coherently, $R \approx 0.879$ false boundary
Pooled offset as specified	[RETIRED 14–15 AUG] multiplicative bias $\mathbb{E}[\cos \varphi]$
Repaired $(\hat{n}, \hat{c}, \hat{t})$: project onto pooled normal, distance from crossing law	[VALIDATED] coverage 0.945–0.998, $5\times$ narrower
Axis dominance as permutation-calibrated significance	[RETIRED 14–15 AUG] null $q_{95} = 0.9969$; fires 0.20–0.79 on oblique gates
Axis dominance as certified equivalence bound	[VALIDATED] tracks true tilt; false claims 0.000 beyond tolerance
Interval coverage on $\hat{t}$	[VALIDATED] (14 Aug) 0.00 as specified → 0.945–0.998 repaired; scope: $\theta = 0$, $d = 2$, by-design placement
The separability guard	UNTESTED — load-bearing; the next experiment